Physik II — FS 2024 — Prof. R. Wallny — Serie 10

Solution

Exercise 1: Proton Beam

a) The total energy is given by

$$E_{\text{tot}} = E_{\text{kin}} + mc^2 = mc^2 + mc^2 = 2mc^2\,. \tag{1}$$

Because E_{tot} can be written also as

$$E_{\text{tot}} = \gamma mc^2, \tag{2}$$

we find that

$$\gamma = \frac{1}{\sqrt{1-\beta^2}} = 2\,. \tag{3}$$

Solving this equation for β yields

$$1-\beta^2 = \frac{1}{\gamma^2} \tag{4}$$

$$\beta = \sqrt{1-\frac{1}{\gamma^2}} = \sqrt{1-\frac{1}{2^2}} = \frac{\sqrt{3}}{2}. \tag{5}$$

Thus,

$$v = c\cdot\beta = \frac{\sqrt{3}}{2}c\,. \tag{6}$$

b) Starting from the definition of current,

$$I = \frac{dQ}{dt}, \tag{7}$$

and expressing the charge Q via the charge density ρ and the volume V:

$$Q = \rho\cdot V \tag{8}$$

we can deduce ρ, ($[\rho] = \frac{\text{C}}{\text{m}^3}$).
More precisely, the current can be expressed as

$$I = \frac{dQ}{dt} \tag{9}$$
$$= \frac{\rho\, dV}{dt} \tag{10}$$
$$= \frac{\rho A\, dx}{dt} \tag{11}$$
$$= \rho A \frac{dx}{dt} \tag{12}$$
$$= \rho \pi R_0^2 v\,. \tag{13}$$

where $A = \pi R_0^2$ is the transverse area of the beam.
Solving the above equation for ρ leads to

$$\rho = \frac{I}{\pi R_0^2 v} = \frac{2\sqrt{3}I}{3\pi R_0^2 c}. \tag{14}$$

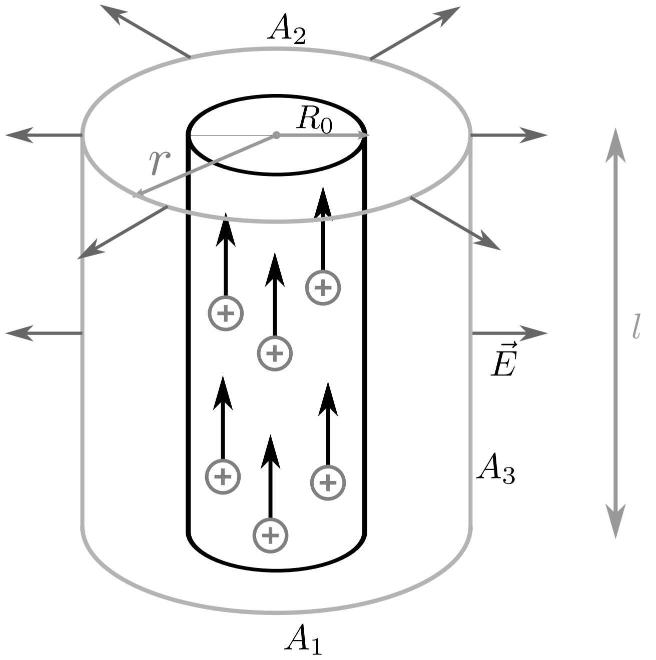


Abbildung 1.1: Electric field of the proton beam in the laboratory system with illustration of the cylindircal symmetry and the integration volume.

c) We are using cylindrical coordinates. From symmetry arguments we know that the electric field produced by a linear charge distribution oriented in the z-direction points in the radial direction

$$\vec{\mathbf{E}} = E(r)\vec{\mathbf{e}}_r. \tag{15}$$

Using Gauss' theorem we can calculate the electric field strength

$$\int_A \vec{\mathbf{E}} \cdot d\vec{\mathbf{A}} = \frac{Q_{\text{eing.}}}{\varepsilon_0} \; . \tag{16}$$

For this purpose we make use of the cylindrical surface around the beam as shown in Figure 1.1. Since the electric field on the two flat, circular surfaces ('Stirnflächen') of the cylinder ($A1$ and $A2$) is orthogonal to the surface normal we have to consider only the curved surface ('Mantelfläche') $A3$ around the cylinder.

$$\int_A \vec{\mathbf{E}} \cdot d\vec{\mathbf{A}} = \underbrace{\int_{A1} \vec{\mathbf{E}} \cdot d\vec{\mathbf{A}}}_{0} + \underbrace{\int_{A2} \vec{\mathbf{E}} \cdot d\vec{\mathbf{A}}}_{0} + \int_{A3} \vec{\mathbf{E}} \cdot d\vec{\mathbf{A}} \tag{17}$$

$$= 2\pi r l E(r) \; . \tag{18}$$

The charge enclosed by the cylinder with radius r is

$$Q_{\text{eing.}} = \begin{cases} \rho\pi r^2 l & r \leq R_0, \\ \rho\pi R_0^2 l & r > R_0. \end{cases} \tag{19}$$

Hence, using equation (16), the electric field reads

$$E(r) = \begin{cases} \frac{\rho r}{2\varepsilon_0} & r \leq R_0, \\ \\ \frac{\rho R_0^2}{2\varepsilon_0 r} & r > R_0. \end{cases} \tag{20}$$

d) The charge density in the proton reference system Σ' is

$$\rho' = \frac{Q'}{V'} = \frac{Q'}{dx'dy'dz'} = \frac{Q}{\gamma dxdydz}. \tag{21}$$

In the last equation we have used the fact that the charge is Lorentz invariant ($Q = Q'$) and that the space-time transformation reading

$$\begin{pmatrix} dx' \\ dy' \\ dz' \\ dt' \end{pmatrix} = \begin{pmatrix} \gamma(dx - vdt) \\ dy \\ dz \\ \gamma(dt - \frac{v}{c^2}dx) \end{pmatrix}. \tag{22}$$

implies that $dx' = \gamma dx$, $dy' = dy$ and $dz' = dz$ Note that the "measurement" of the volume is effectuated instantaneously, i.e. with $dt = 0$ so that $dx' = \gamma dx$.

Hence,

$$\rho' = \frac{Q}{\gamma dxdydz} = \frac{Q}{\gamma V} = \frac{\rho}{\gamma}. \tag{23}$$

e) As a consequence, the electric field in the proton reference frame can be calculated using a similar approach as used to deduce Equation (20).

$$\vec{\mathbf{E}}' = \begin{cases} \frac{\rho r'}{2\varepsilon_0}\frac{\vec{\mathbf{e}}_{r'}}{\gamma} & r' \le R_0', \\ \\ \frac{\rho R_0'^2}{2\varepsilon_0 r'}\frac{\vec{\mathbf{e}}_{r'}}{\gamma} & r' > R_0'. \end{cases} \tag{24}$$

Because the transverse directions are not affected by contraction $r' = r$, $R_0' = R_0$ and $\vec{\mathbf{e}}_{r'} = \vec{\mathbf{e}}_r$ the above equation simplifies to

$$\vec{\mathbf{E}}' = \frac{1}{\gamma}\vec{\mathbf{E}} \tag{25}$$

ALTERNATIVELY:
Knowing that there are no moving charges in the proton frame, one can assume that there are no magnetic fields $\vec{\mathbf{B}}' = 0$. Thus the electric field $\vec{\mathbf{E}}_\perp$ in the Lab frame perpendicular to the beam can be calculated from the one in the proton frame by

$$\vec{\mathbf{E}}_\perp = \gamma \vec{\mathbf{E}}'_\perp \tag{26}$$

$$\vec{\mathbf{E}}_\parallel = \vec{\mathbf{E}}'_\parallel = 0 \tag{27}$$

This equation can only be applied, if no magnetic fields are present. As the Lab frame field $\vec{\mathbf{E}}$ is only perpendicular to the beam direction,

$$\vec{\mathbf{E}}' = \vec{\mathbf{E}}'_\perp = \frac{1}{\gamma}\vec{\mathbf{E}}_\perp = \frac{1}{\gamma}\vec{\mathbf{E}} = \frac{\rho}{2\gamma\varepsilon_0}\vec{\mathbf{e}}_r \begin{cases} r & r \le R_0 \\ \frac{R_0^2}{r} & r > R_0 \end{cases} \tag{28}$$

is obtained. Note that $r = r'$ as it is perpendicular to the beam and therefore not subject to Lorentz contractions.

f) In the proton reference frame as the charges are non moving, there is only the radially directed electric field to consider (the magnetic field is zero $B' = 0$). Consequently, the acceleration at the radial position $r' = r$ points in radial direction and amounts to

$$a' = \frac{d^2r'}{dt'^2} = \frac{F'}{m} = \frac{qE'}{m} = \frac{q\rho r}{2\varepsilon_0 m\gamma}, \tag{29}$$

where q is the proton charge. Note that q and m are Lorentz invariants, i.e. $q = q'$ and $m = m'$. The transform to the lab frame can be obtained simply by applying the relation $dr' = dr$, $dt' = \frac{dt}{\gamma}$ (note that the time in the co-moving system is always the shortest) to obtain

$$dr = \frac{q\rho r}{4\varepsilon_0 m\gamma}\left(\frac{dt}{\gamma}\right)^2. \tag{30}$$

From this, the acceleration in the lab frame is found to be:

$$a = \frac{\mathrm{d}^2 r}{\mathrm{d}t^2} = \frac{q\rho r}{2\varepsilon_0 m\gamma^3} = \frac{q\rho r}{16\varepsilon_0 m}. \tag{31}$$

In the last equation we have used the fact that $\gamma = 2$.

Inserting ρ from Eq. (14) in the last equation, one obtains the acceleration in the lab frame

$$a = \frac{\sqrt{3} I q r}{24\varepsilon_0 \pi m c R_0^2} \tag{32}$$

which points in radial direction as in the proton frame.

Exercise 2: Ampère's law

In order to solve this problem we will use Ampere's law, which relates the line integral of the magnetic field along the boundary of a surface to the current that passes through this surface. Because of symmetry reasons we know that the magnetic field is tangential to the boundary ∂C and its magnitude there is constant. The sign of the field is given by the "right-hand-rule". Let's consider the first case (i). For a circular surface C coaxial with the cylinders of the problem and with radius $r < a$ we find

$$\oint_{\partial C} \vec{B} \cdot d\vec{l} = B \oint_{\partial C} dl = B 2\pi r = \mu_0 \int_C \vec{J} \cdot d\vec{S}. \tag{33}$$

The integral of the current density is the total amount of current through the circle:

$$\int_C \vec{J} \cdot d\vec{S} = |\vec{J}_\mathrm{i}| \int_C dS = \frac{I}{\pi a^2}\pi r^2 = I\left(\frac{r}{a}\right)^2 \tag{34}$$

and therefore

$$B_\mathrm{i} = \mu_0 \frac{Ir}{2\pi a^2}. \tag{35}$$

In the other cases we will increase the radius of our circle and apply Ampere's law again. On the left side of the equation we will always have the same term:

$$\oint_{\partial C} \vec{B} \cdot d\vec{l} = B \oint_{\partial C} dl = B 2\pi r = \mu_0 \int_C \vec{J} \cdot d\vec{S}. \tag{36}$$

However, the current flowing through the circle will be different in each of the cases. For the situation (ii) the whole inner conductor is within the circle so $\int_C \vec{J} \cdot d\vec{S} = I$ and therefore

$$B_\mathrm{ii} = \mu_0 \frac{I}{2\pi r}. \tag{37}$$

In the case (iii) the whole inner conductor is within the circle, but now we also have an opposite contribution coming from the outer conductor. The integral of the current density is therefore:

$$\int_C \vec{J}\cdot d\vec{S} = I - |\vec{J}_{\rm iii}|\pi(r^2-b^2) = I\left(1-\frac{r^2-b^2}{c^2-b^2}\right) = I\frac{c^2-r^2}{c^2-b^2} \tag{38}$$

and the magnetic field

$$B_{\rm iii} = \mu_0\frac{I}{2\pi r}\frac{c^2-r^2}{c^2-b^2}. \tag{39}$$

Finally, in the case (iv) the current flowing through the circle is:

$$\int_C \vec{J}\cdot d\vec{S} = I - I = 0 \tag{40}$$

and consequently

$$B_{\rm iv} = 0. \tag{41}$$

Exercise 3: Charged particles in electric and magnetic fields.

Starting from the expression for the Lorentz force acting on a charged particle in an arbitrary electromagnetic field, the equation of motion of a charged particle can be written as:

$$m\dot{\vec{v}} = \vec{F}_{Lorentz} = q\left(\vec{E}+\vec{v}\times\vec{B}\right) \tag{42}$$

a) In the region of space between the two plates P1 and P2 the charged particle is accelerated by the electric field, and its kinetic energy is increased at the expense of the work done by this electric field. The increase of the kinetic energy of a particle transversing the potential difference ΔV (and which was initially at rest) is $\frac{1}{2}mv^2 - 0 = |q\Delta V|$ and the velocity of the particle entering the region of space MS is:

$$v_0 = \sqrt{2q\Delta V/m} \tag{43}$$

b) Taking into account the configuration of fields in the MS region, the equation of motion (42) can be decomposed in three components:

$$m\dot{v}_x = 0 \tag{44}$$
$$m\dot{v}_y = 0 + qBv_z \tag{45}$$
$$m\dot{v}_z = qE - qBv_y \tag{46}$$

Since the initial velocity in the x-direction is zero, the motion of the particle will be constrained to the yz-plane. The system of two differential equations (45) and (46) can be solved by inserting equation (46) into the derivative of equation (45). In this way we obtain a non-homogeneous linear differential equation of second order with constant coefficients:

$$\ddot{v}_y + \left(\frac{qB}{m}\right)^2 v_y = \left(\frac{qB}{m}\right)^2\frac{E}{B} \tag{47}$$

The solution of the diff. equation (47) is a sum of the homogeneous and a particular solution of this equation: $v_y(t) = v_y^{(h)}(t) + v_y^{(p)}(t)$. The homogeneous solution of this differential equation is of the form:

$$v_y^{(h)}(t) = C_1\cos\Omega t + C_2\sin\Omega t$$

where $\Omega = \frac{qB}{m}$. The particular solution is of the form $v_y^{(p)}(t) = const$. Therefore, $\dot{v}_y^{(p)} = 0$ and using equation (45) $v_z^{(p)} = \dot{v}_z^{(p)} = 0$. From equation (46) it follows that

$$v_y^{(p)}(t) = \frac{E}{B} \tag{48}$$

For this reason, the complete solution of the diff. equation (47) is of the form:

$$v_y(t) = C_1 \cos \Omega t + C_2 \sin \Omega t + \frac{E}{B} \tag{49}$$

Constants C_1 and C_2 can be determined from initial conditions. At the time $t = 0$ the particle has entered the MS region with velocity $\vec{v}(t = 0) = (0, v_0, 0)$. Therefore, according to equations (49) and (45), we obtain the values for constants $C_1 = v_0 - \frac{E}{B}$ and $C_2 = 0$. The full solution of equations of motion for our charged particle is:

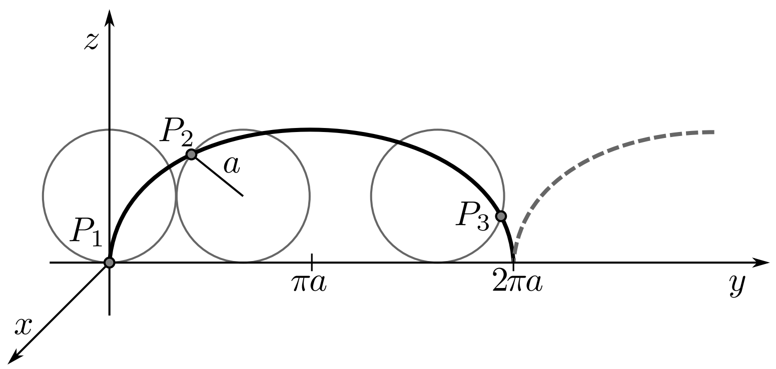
$$\begin{aligned} v_y(t) &= \frac{E}{B} - \left(\frac{E}{B} - v_0\right) \cos \Omega t \\ v_z(t) &= \left(\frac{E}{B} - v_0\right) \sin \Omega t \end{aligned}$$

The trajectory of the charged particle can be obtained by integrating the previous equations and including initial conditions. The trajectory is described by the following set of equations:

$$y(t) = \frac{E}{B}t - \Omega^{-1}\left(\frac{E}{B} - v_0\right) \sin \Omega t \tag{50}$$

$$z(t) = \Omega^{-1}\left(\frac{E}{B} - v_0\right)(1 - \cos \Omega t) \tag{51}$$

In case of $v_0 = 0$, these equations describe a cycloid - the trajectory of a point fixed on the rim of a wheel with radius $a = \frac{E}{B\Omega}$ which rolls with the velocity $\frac{E}{B}$ (see figure below).



c) According to equations (50) and (51), the particle will move in a straight line in the MS region if the factor $\frac{E}{B} - v_0$ vanishes. In order to achieve this, the intensities of the electric and magnetic field in the MS region have to be adjusted so that $E = Bv_0$. Using this condition and the equation (43) we can determine the specific charge of our charged particle:

$$\frac{q}{m} = \frac{E^2}{2\Delta V B^2}$$

This method can be used to measure the specific charge of particles.

d) If the electrical field in the MS region is switched off ($E = 0$), the equations of motion (50) and (51) reduce to the equations of circular motion with radius $r = \frac{v_0}{\Omega} = \frac{mv_0}{qB}$, and therefore, taking into account the equation (43):

$$r = \sqrt{\frac{2m\Delta V}{qB^2}} \tag{52}$$

We see that the radius of a circular trajectory depends on the mass and charge of the accelerated particle. This dependence is, for example, used in Mass-spectrometers to differentiate particles with the same charge but with different masses.
If the particle detector in the MS region detects hits of particles 1 and 2 at positions $-z_1$ and $-z_2$, we can conclude that the radius of trajectories of detected particles were $r_1 = \frac{z_1}{2}$ and $r_2 = \frac{z_2}{2}$ respectively. According to equation (52), we can find the difference between the masses of particles detected at positions z_1 and z_2:

$$\Delta m = m_2 - m_1 = \frac{qB^2}{8\Delta V}(z_2^2 - z_1^2)$$

Exercise 4: Cyclotron

a) To keep the protons on their orbit, the Lorentz force has to act as centripetal force. Since the protons always move perpendicular to the magnetic field, we can set the the magnitude of these forces equal to each other:

$$F_L = evB = m_p \frac{v^2}{r} = F_Z, \tag{53}$$

with m_p the proton mass, e the elementary charge, r the (current) radius of the proton orbit and v the (current) proton velocity. The maximum possible radius of the orbit is $r_{\text{max}} = D/2$. If we put that in, we get:

$$eB = \frac{m_p v_{\text{max}}}{r_{\text{max}}} \tag{54}$$

and the maximal velocity:

$$v_{\text{max}} = \frac{eBD}{2m_p}. \tag{55}$$

Therefore the kinetic energy at the exit is

$$E_{\text{kin,max}} = \frac{1}{2} m_p v_{\text{max}}^2 = \frac{(eBD)^2}{8m_p}. \tag{56}$$

b) Also from the fact that the Lorentz force acts as centripetal force, one can derive the frequency. For a given fixed kinetic energy (and velocity), the protons are on a circular orbit, with frequency $f = \frac{\omega}{2\pi} = \frac{v}{2\pi r}$. Now we put in $eB = \frac{m_p v}{r}$ respectively $v = \frac{eBr}{m_p}$ from part a), so we get:

$$f = \frac{eB}{2\pi m_p}. \tag{57}$$

As we can see, this frequency (called cyclotron frequency) is independent of the energy (respectively the velocity) of the protons and of the radius of the orbit.

c) Whenever a proton passes the gap, its kinetic energy is increased by eV. Therefore the number of turns is given by:

$$N = \frac{E_{\text{kin,max}}}{2eV} = \frac{e(BD)^2}{16m_pV}. \tag{58}$$

The factor of two in the denominator comes from the fact, that the proton passes the acceleration gap twice per turn. Therefore the time a proton spends in the cyclotron is

$$t_0 = \frac{N}{f} = \frac{\pi BD^2}{8V}, \tag{59}$$

independent of the proton mass and charge.

Exercise 5: Biot-Savart "around the corner"

a) The law of Biot-Savart states this general formula:

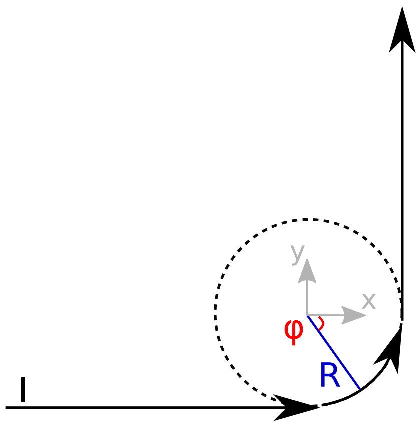
$$\vec{B}(\vec{r}) = \frac{\mu_0 I}{4\pi}\int_C \mathrm{d}s \frac{\hat{e}_I \times (\vec{r} - \vec{r}_0(s))}{|\vec{r} - \vec{r}_0(s)|^3} \ . \tag{60}$$

Here, $\hat{e}_I$ denotes the current direction, $\vec{r}$ the vector to the point at which the magnetic field is to be determined and $\vec{r}_0(s)$ the parametrization of the curve.

b) We choose the coordinate system such, that its origin lies in the center of the curve. Therefore it follows that

$$\vec{r} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \tag{61}$$

is the point at which the magnetic field is to be determined.



In order to resolve the individual components, we use $\hat{e}_I$ and $\vec{r}_0$ for the three sub-areas "$\rightarrow$",

"$\circlearrowleft$" and "$\uparrow$". The following applies

$$\text{“}\rightarrow\text{“}: \qquad \hat{e}_I = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \qquad \vec{r}_0 = \begin{pmatrix} x \\ -R \\ 0 \end{pmatrix} \tag{62}$$

$$\text{“}\circlearrowleft\text{“}: \qquad \hat{e}_I = \begin{pmatrix} -\sin\varphi \\ \cos\varphi \\ 0 \end{pmatrix} \qquad \vec{r}_0 = \begin{pmatrix} R\cos\varphi \\ R\sin\varphi \\ 0 \end{pmatrix} \tag{63}$$

$$\text{“}\uparrow\text{“}: \qquad \hat{e}_I = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \qquad \vec{r}_0 = \begin{pmatrix} R \\ y \\ 0 \end{pmatrix} \tag{64}$$

Therefore we get

$$\vec{B}(\vec{r}=0) = \vec{B}_{\rightarrow}(\vec{r}=0) + \vec{B}_{\circlearrowleft}(\vec{r}=0) + \vec{B}_{\uparrow}(\vec{r}=0) \tag{65}$$

$$\begin{aligned} &= \frac{\mu_0 I}{4\pi}\int_{-\infty}^{0} \mathrm{d}x \frac{\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \times \begin{pmatrix} -x \\ R \\ 0 \end{pmatrix}}{\left|\begin{pmatrix} x \\ -R \\ 0 \end{pmatrix}\right|^3} \\ &+ \frac{\mu_0 I}{4\pi}\int_{-\pi/2}^{0} R\mathrm{d}\varphi \frac{\begin{pmatrix} -\sin\varphi \\ \cos\varphi \\ 0 \end{pmatrix} \times \begin{pmatrix} -R\cos\varphi \\ -R\sin\varphi \\ 0 \end{pmatrix}}{\left|\begin{pmatrix} R\cos\varphi \\ R\sin\varphi \\ 0 \end{pmatrix}\right|^3} \\ &+ \frac{\mu_0 I}{4\pi}\int_{0}^{\infty} \mathrm{d}y \frac{\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \times \begin{pmatrix} -R \\ -y \\ 0 \end{pmatrix}}{\left|\begin{pmatrix} R \\ y \\ 0 \end{pmatrix}\right|^3} \end{aligned} \tag{66}$$

c) In the next step, we execute the integration of the single terms explicitly. This gives us

$$\vec{B}_{\rightarrow}(\vec{r}=0) = \frac{\mu_0 I}{4\pi}\int_{-\infty}^{0} \mathrm{d}x \frac{\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \times \begin{pmatrix} -x \\ R \\ 0 \end{pmatrix}}{\left|\begin{pmatrix} x \\ -R \\ 0 \end{pmatrix}\right|^3} \tag{67}$$

$$= \frac{\mu_0 I}{4\pi}\int_{-\infty}^{0} \mathrm{d}x \frac{R\hat{e}_z}{(x^2+R^2)^{3/2}} \tag{68}$$

$$= \frac{\mu_0 I}{4\pi}\frac{\hat{e}_z}{R}\,, \tag{69}$$

where $\hat{e}_z$ is the unit vector in z-direction. Analogously, we get the contribution for the second straight part

$$\vec{B}_{\uparrow}(\vec{r}=0) = \frac{\mu_0 I}{4\pi} \int_0^{\infty} \mathrm{d}y \frac{\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \times \begin{pmatrix} -R \\ -y \\ 0 \end{pmatrix}}{\left| \begin{pmatrix} R \\ y \\ 0 \end{pmatrix} \right|^3} \tag{70}$$

$$= \frac{\mu_0 I}{4\pi} \frac{\hat{e}_z}{R} \ . \tag{71}$$

The integration over the curved part results in

$$\vec{B}_{\circlearrowleft}(\vec{r}=0) = \frac{\mu_0 I}{4\pi} \int_{-\pi/2}^{0} R\mathrm{d}\varphi \frac{\begin{pmatrix} -\sin\varphi \\ \cos\varphi \\ 0 \end{pmatrix} \times \begin{pmatrix} -R\cos\varphi \\ -R\sin\varphi \\ 0 \end{pmatrix}}{\left| \begin{pmatrix} R\cos\varphi \\ R\sin\varphi \\ 0 \end{pmatrix} \right|^3} \tag{72}$$

$$= \frac{\mu_0 I}{4\pi} \int_{-\pi/2}^{0} R\mathrm{d}\varphi \frac{R\hat{e}_z}{R^3} \tag{73}$$

$$= \frac{\mu_0 I}{4\pi} \frac{\pi \hat{e}_z}{2R} \ . \tag{74}$$

d) In total we get therefore the magnetic field at the point $\vec{r}$ at the center of the curve

$$\vec{B}(\vec{r}=0) = \frac{\mu_0 I}{4\pi} \left(\frac{2}{R} + \frac{\pi}{2R} \right) \hat{e}_z \ , \tag{75}$$

which is, as expected, exactly two times the magnetic fiels of a 1/2 infinite wire plus 1/4 the magnetic field of a circular loop.